Virtual reality (VR) and augmented reality (AR): While the metaverse is a platform that is still developing, virtual reality is a technology that is already being used in daily lives. (The metaverse may even replace the internet as we know it today.) Virtual reality can be experienced today in diverse real-time applications like simulators, gaming, Industry 4.0, etc. Metaverse can be experienced via VR and AR. Metaverse technology is far more advanced than virtual reality and is expanding to 3D depiction of the internet or virtual world.

Blockchain and cryptocurrency: 'Crypto metaverses' have a huge social and financial potential, and enable the integration of blockchain infrastructure with the metaverse virtual world. With the fusion of virtual environments using VR, games, social media networking and crypto exchanges, the metaverse could become a 'central element' for the next level of blockchain gaming.

Crypto metaverse is defined as a metaverse that integrates blockchain technology and crypto assets such as 'metaverse tokens'. Examples of crypto metaverses are: Decentraland, The Sandbox, Aliens World and Cryptovoxels. With VR, blockchain, and crypto, the gaming industry can experience a revolution that can be called 'blockchain gaming'. Metaverse crypto assets include digital objects, land, etc, details of which can be recorded on a blockchain and even traded using Bitcoins or Ether on diverse decentralised exchanges (DEXs).

Key features:

- *Decentralisation:* Crypto metaverses are decentralised, giving all metaverse games developed on blockchain technology strong value models.
- *Provable provenance:* Crypto tokens called non-fungible tokens (NFTs) will enable transparency and access to asset markets in the gaming industry.
- *Real-world economic value:* Metaverse tokens can be exchanged on DEXs and NFT marketplaces.

Artificial intelligence: Fusion of artificial intelligence (AI) with the metaverse adds value to infrastructure, and gives better information to all the top layers of the metaverse.

Metaverse makes use of AR, VR, AI and blockchain to make advanced replicas of the real world in a virtual world. It applies advanced AI algorithms to perform content analysis, speech processing, computer vision, and more.

This is how the metaverse uses AI.

Virtual avatars: With the use of AI, 2D or 3D user images can be scanned and transformed to very realistic *avatars.*

Digital humans: Digital humans have human-like capabilities for seeing, listening and understanding the conversation of real humans. They can use digital speech and body language to create human-like conversations. In the metaverse, digital humans can be considered as 3D chatbots that respond just like human beings. They are called 'non-playing characters' whose actions and outcomes are measured by automated scripts or sets of rules.

Language processing: With AI, all languages can be transformed to a machine readable format that is analysed, and the output is generated back in the same language. By using advanced ML and DL (deep learning) algorithms, language translation becomes more accurate and fast in response.

Data learning: AI helps generate models into which historical data is fed. DL and ML are used to generate new outputs, and as more and more data is fed into the model, the outputs get better.

3D graphics: Metaverse is also dependent on 3D technologies. 3D graphics generate images or objects using 3D graphics software. These can be viewed using specialised hardware called 3D displays, and have diverse applications in the real world.

3D reconstruction is a technology that uses 3D models to visualise an object or environment in a three-dimensional view. The metaverse can only exist within the three-dimensional setting. It's essentially a digital twin of our world.

Meta has rolled out diverse devices like Oculus VR to experience the metaverse.

The Internet of Things: IoT technology provides a strong bridge between the physical and virtual worlds. It helps analyse data from the physical world using smart sensors. IoT and the metaverse can together take smart systems to the next level.

Metaverse can leverage IoT in the following ways.

'Real' feeling in devices: Via IoT, the metaverse can make virtual devices look real, and even solve complex problems of hybrid data integration with cloud and digital infrastructures.

Smart integration: IoT can help the metaverse collaborate with the real world. It is a strong building block for creating interoperable systems and fusing digital content into the physical environment.

Data collection: IoT enables digital twins and virtual simulations in the metaverse. It helps integrate scanned objects with real metadata from the physical environment.

Universal scene description (USD) for metaverse development

When there are multiple metaverse platforms, there are no uniform technology solutions on how to develop the metaverse platform and the 3D objects in it. Pixar was using universal scene description (USD) for a long time to describe 3D objects, and open sourced it in 2016. Now, for metaverse platform development, many companies like Adobe, NVIDIA, Siemens and Lowe's have started looking into USD as the standard solution for metaverse-based 3D object rendering and development.

Initially, when Pixar used USD, it did so mainly for media related image rendering in the 3D space, in the entertainment sector. We can now expect more real-time use cases across different sectors like medical and healthcare, robotics and retail. USD is more than animation and simulation of